paraffin is bound to reduce the effect, a quite considerable intensification of the latter is the result. This method for the increase of the output in nuclear transmutations by means of slowing down neutrons, is employed very frequently in nuclear physics.

## II. THE PROCEDURES FOR PRODUCING NUCLEAR TRANSMUTATION

As a general rule, particles very rich in energy are required to produce a nuclear transmutation. It is only when the transmutation is induced by neutrons that the energy content of the bombarding particle is frequently reduced deliberately as much as possible. But neutrons, to start with, must be produced by a nuclear reaction induced by fast particles, such, for instance, as the bombardment of beryllium by alpha particles.

Nature itself provides us with the most convenient source of energy-rich particles: The natural alpha emitters. To be sure, the radiation of even the most powerful radioactive preparations is always relatively weak, and sufficient for the transmutation of only a small number of atoms. On the other hand, in addition to the alpha particles, still other types of particles, namely, fast protons and deuterons, are needed in order to produce all possible sorts of nuclear transformation.

The most logical method of producing particles of high energy consists in a very strong acceleration of charged particles by a very high voltage, possibly 1,000,000 volts or more. Of course, direct-current voltage must be used. It is of course much more difficult to produce such a high direct-current voltage than it would be to generate an alternating-current voltage of the same magnitude.

The system known as the *Greinacher circuit* (Figure 33) is one very frequently employed today in high-voltage generators. Two columns, with condensers, *C,* are connected by a system of valves, *V,* each of which permits a beam of electrons to pass through in one direction only, for instance, in the direction indicated by arrows in our diagram. The entire system is designed so that the point *d,* for instance, can be charged positively (but not negatively) relative to the earthed point *a,* without being discharged. Similarly, *c* can be charged positively relative to *d,* likewise *f* relative to *c, e* relative *to f,* etc. Now, if an alternating-current voltage (usually between 200 and 300 kv or so) is applied, by means of a transformer, *T,* between the two columns, the points *d, c, f,* etc., become positively charged through the valves until the alternating-current voltage of, for instance, the point *d* never falls, throughout an entire period, below that of *a,* which may be taken as equal to 0; for otherwise a current would still flow through the valve $V_1$. Thus, if the peak